Controlling spectral centrality measures: PageRank case study

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Introduction and motivation

Any system with elements and interactions or connections between them can be modelled as a complex network. Network-theoretic methods can then be applied to extract useful information about the system.

One of the key features of a network is its *centrality*: the importance of each node in the network with respect to a given measure. The celebrated PageRank algorithm [1], computing the stationary distribution $\pi(\alpha, v)$ of a certain Markov chain given by

$$\pi^T = \pi^T(\alpha P + (1-\alpha)ev^T), \tag{1}$$

is one of the most relevant ones, and as such, understanding its outcomes is an important task.

Main Problem: Centrality control

An interesting question once we are given a network and a centrality measure is whether any possible centrality outcome can be realized after suitable changes.

In [2] it was shown that another centrality measure, the eigenvector centrality, can be completely specified if one is able to tune the weights of around 10% of the network's connections.

In the PageRank case, the normalization of the adjacency matrix renders the weight-tuning useless. But we can instead consider tuning the two parameters α (damping factor) and v (personalization vector), in the hopes of obtaining any desired outcome.

PageRank - full control

First we address the question of whether we can achieve any desired centrality $\pi(\alpha, v)$ changing its two parameters.

We proved the following existence theorem [3].

Theorem 1. *Given $G = (V, E)$ and $\pi_0 \in \mathbb{R}^n_+$ with $|\pi_0|_1 = 1$, then there exists $v \in \mathbb{R}^n_+$ with $|v|_1 = 1$ personalization vector such that $\pi(G, \alpha, v) = \pi_0$ if and only if $\langle \pi_0, e_j \rangle > \alpha \pi_0^T P e_j$ for all $1 \leq j \leq n$.*

This is very restrictive: in general, the damping factor α will need to be incredibly small in order to fulfill the n inequalities. The answer to the original problem is that no, in practice there can't be full external control.

Can we relax the original centrality control problem, to allow for more controllability?

PageRank - ranking control

For most applications, one is not concerned with the specific values of the centrality vector. One is interested in the *ranking*, i.e. the ordering of importance.

It can be shown [3] that for any ranking to be possible, the uniform centrality $\pi = (1, ..., 1)/n$ needs to be possible. This, along with Theorem 1, allows us to state the following theorem:

Theorem 2. *Given a network $G = (V, E)$ and $\alpha \in (0, 1)$, then it is possible to obtain any ranking of the nodes under the PageRank $\pi(G, \alpha, v)$ for a personalization vector $v \in \mathbb{R}^n_+$ with $|v|_1 = 1$ if and only if*

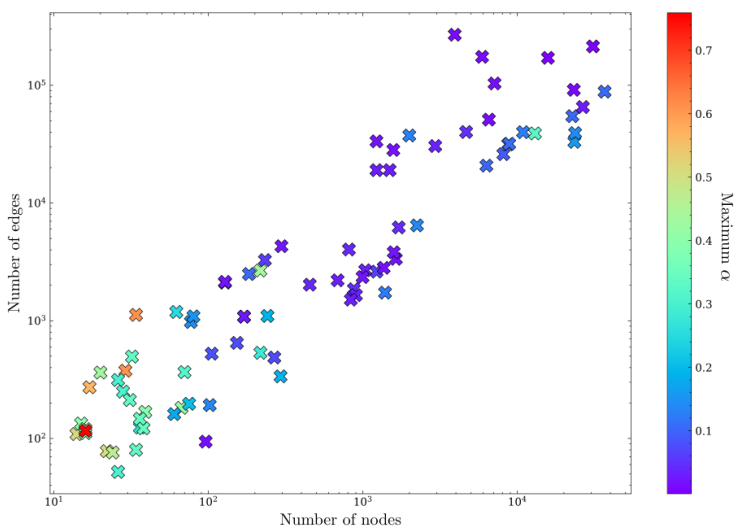
$$\frac{1}{\alpha} > \max_j \left(\sum_{i=1}^{n} P_{ij}\right). \tag{2}$$

Computation over real network datasets

We computed the maximum α necessary to achieve ranking control over the centrality of real network datasets. This node-edge scatter plot represents with a color bar the maximum α that each of them allows.

We found that, even in the less restrictive case of ranking control, we are only able to use $\alpha \sim 0.1$, while for real applications (e.g. the Google search engine) it has a value around 0.85 [4].

This result proves, once again, the fact that PageRank is a remarkably stable centrality measure.



References

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